

Bayesian quantile correction and synthesis for climate products

Antonio Aguirre Raquel Prado Bruno Sansó
Department of Statistics, University of California, Santa Cruz • jaguir@ucsc.edu

QUESTION

Fast correction before synthesis

Can historical product archives learn source discrepancies, propagate those corrections into new quantile-level ensemble forecasts, and synthesize calibrated posterior predictions fast enough for real-time product updates?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge on the $\log(1 + x)$ scale. Before each origin, USGS flow and retrospective GloFAS/NWS products learn dynamic source discrepancies relative to the gauge. After the origin, issued ensembles and exogenous covariates carry those corrections into source-adjusted quantile forecasts and one posterior predictive flow distribution.



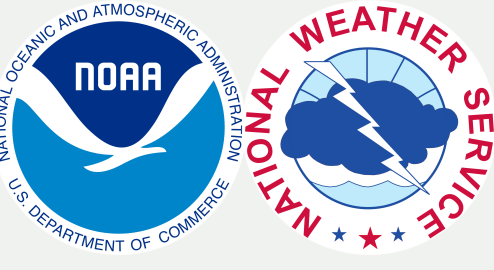
USGS
science for a changing world



ECMWF



GloFAS
Global Flood Awareness System



NOAA
NATIONAL WEATHER SERVICE
U.S. DEPARTMENT OF COMMERCE

USGS observations
15-min public gauge data; daily log-flow target for state updating and held-out verification.

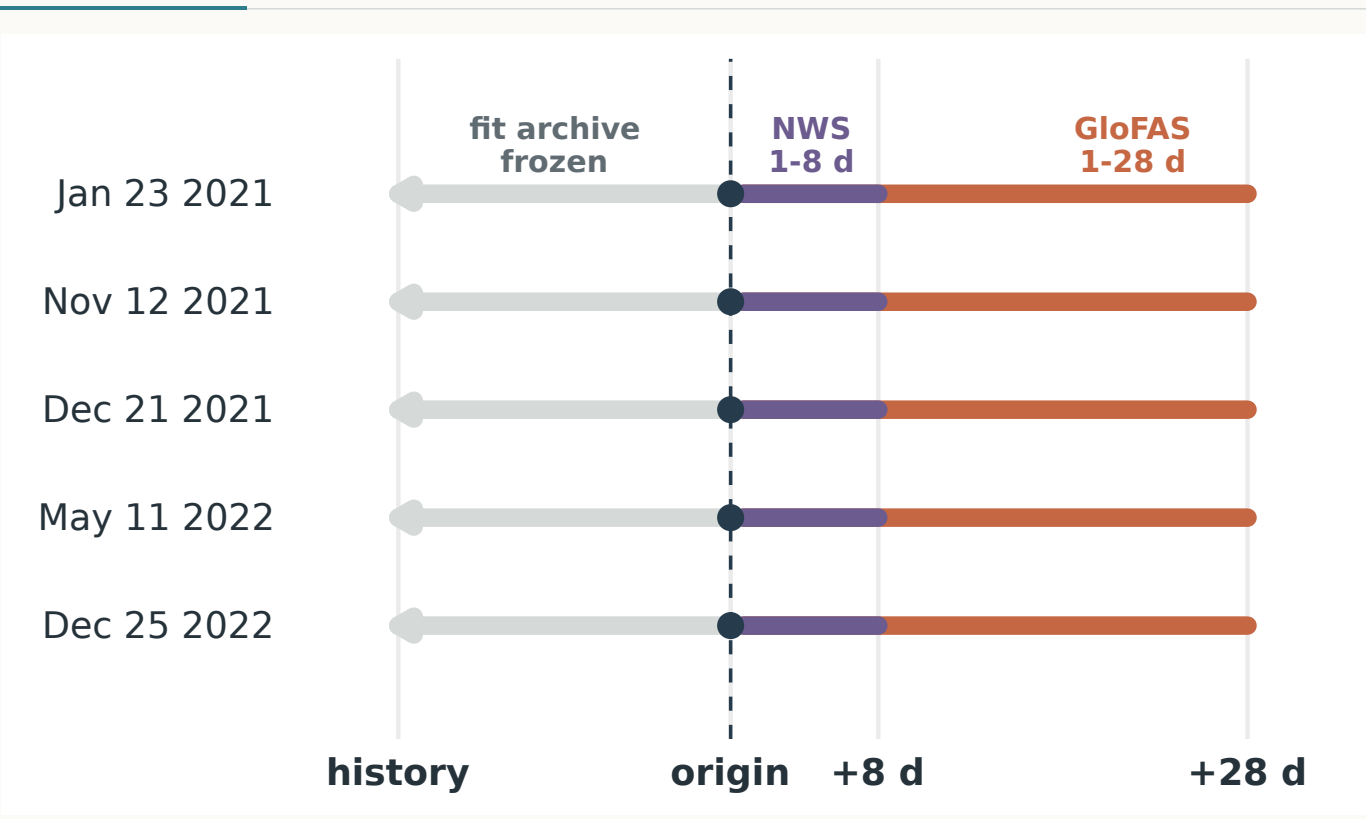
ECMWF GloFAS
retrospective alignment; daily 51-member forecast guidance to 28 days.

NOAA NWS
retrospective alignment; hourly forecasts evaluated on common daily leads 1–8.

Exogenous information
Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

VALIDATION

Rolling-origin holdout design



Strict temporal holdout. At each origin, fitting uses only USGS flow and retrospective products available through that date. Issued forecasts and forecast-window exogenous inputs enter after the origin; future USGS observations are reserved for scoring. Main CRPS uses GloFAS-supported days 1–28, while NWS is compared over common leads 1–8.

INFERENCE

Scalable Bayesian quantile fitting

Seven independent extended Dynamic Quantile Linear Model (exDQLM) lanes target seven levels: 0.05, 0.20, 0.35, 0.50, 0.65, 0.80, and 0.95. The fitted quantile forecasts are checked for crossing in the application and combined into the posterior predictive distribution scored by CRPS.

7 quantile lanes **<12,995** USGS archive days **~2–4 h** fit + synthesis

Parallel quantile lanes

Each quantile level is fit as its own state-space model; the lanes can run concurrently before forecast quantiles are synthesized.

Conjugate DQLM baseline

The asymmetric-Laplace DQLM is fully conjugate under its latent representation; this is the baseline labeled DQLM in the Results panel.

Non-conjugate exDQLM updates

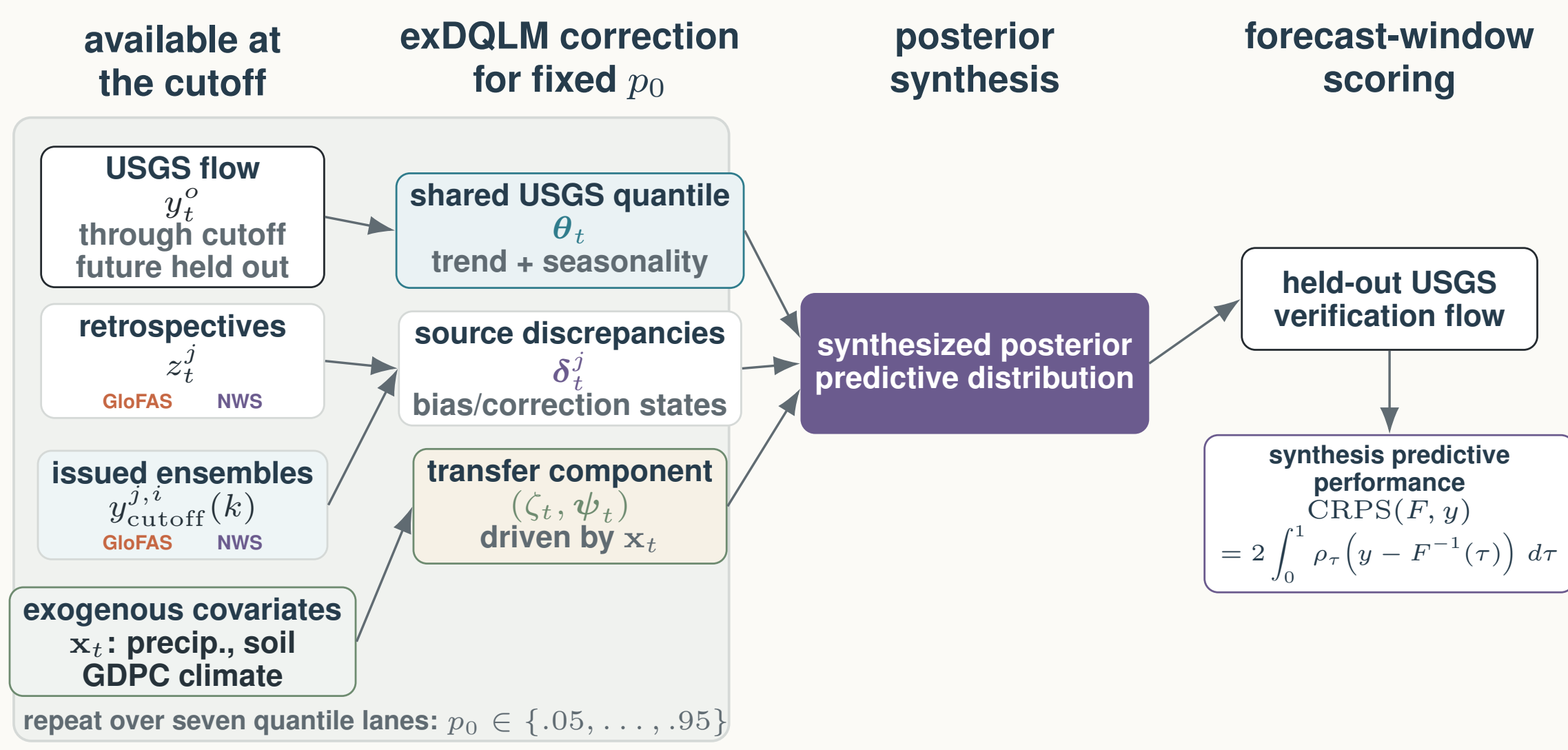
The selected exDQLM replaces AL by exAL. Mean-field variational Bayes keeps Kalman filtering and backward smoothing for Gaussian state blocks, while VB-LD uses local modes, curvature, and Delta-method moments for non-conjugate exAL scale and skewness terms.

State evolution

Component discounting controls trend, multi-period seasonality, source corrections, and exogenous-covariate adjustments.

MODEL

Main Workflow



Source-aware exDQLM workflow. For each target quantile p_0 , pre-origin USGS observations and retrospective product histories identify a latent river-flow quantile and source-specific correction states. After the cutoff, issued ensemble forecasts and staged exogenous covariates propagate corrected forecast-window quantiles. The seven fitted quantile lanes are then synthesized into one posterior predictive distribution and scored only with held-out USGS observations.

extended Dynamic Quantile Linear Model

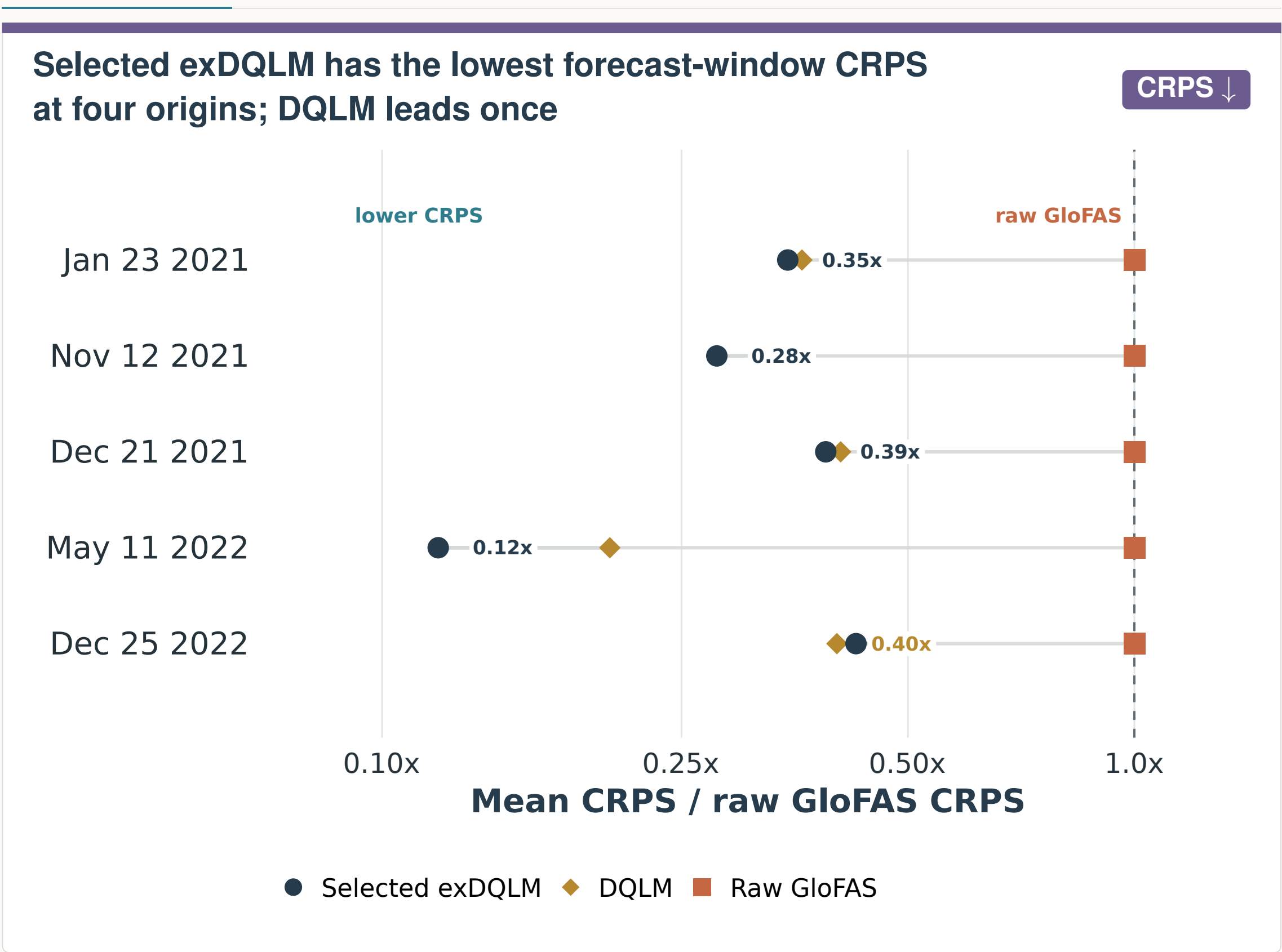
For each target quantile p_0 , let T denote the forecast origin/cutoff. The extended Dynamic Quantile Linear Model (exDQLM) links observed flow, retrospective product histories, and issued forecast members through one latent river-flow quantile. Product-family discrepancies learn source corrections before the origin; the same source channels, with forecast-window covariates x_{T+k} , then define corrected predictive quantiles after the origin.

Observed measurements: $y_t^o \sim \text{exAL}_{p_0}(\mathbf{F}_t^o \theta_t + \zeta_t, \sigma^o, \gamma^o)$, $1 \leq t \leq T$;
Retrospective products: $z_t^r \sim \text{exAL}_{p_0}(\mathbf{F}_t^r(\theta_t + \delta_t^j) + \zeta_t, \sigma^j, \gamma^j)$, $j \in \mathcal{J}$, $1 \leq t \leq T$;
Forecast products: $y_{T+k}^f(k) \sim \text{exAL}_{p_0}(\mathbf{F}_{T+k}^f(\theta_{T+k} + \delta_{T+k}^j + \zeta_{T+k}, \sigma^j, \gamma^j)$, $j \in \mathcal{J}$, $i \in \mathcal{I}_j$, $k \in \mathcal{K}_j$;
Trend and seasonal states: $\theta_t | \theta_{t-1} \sim \mathcal{N}(\mathbf{G}_t \theta_{t-1}, \mathbf{W}_t^0)$, $1 \leq t \leq T + K_{\max}$;
Source correction states: $\delta_t^j | \delta_{t-1}^j \sim \mathcal{N}(\mathbf{G}_t \delta_{t-1}^j, \mathbf{W}_t^j)$, $j \in \mathcal{J}$, $1 \leq t \leq T + K_{\max}$;
Exogenous adjustment: $\zeta_t | \zeta_{t-1}, \psi_{t-1} \sim \mathcal{N}(\lambda \zeta_{t-1} + \mathbf{x}_t^* \psi_{t-1}, \mathbf{W}_t^\zeta)$, $1 \leq t \leq T + K_{\max}$;
Covariate-adjustment states: $\psi_t | \psi_{t-1} \sim \mathcal{N}(\psi_{t-1}, \mathbf{W}_t^\psi)$, $1 \leq t \leq T + K_{\max}$.

exAL_{p_0} is the extended asymmetric Laplace working likelihood for target quantile p_0 ; σ and γ are source-specific scale and skewness terms. T is the forecast origin, j a product family, i an ensemble member, k a lead, and $K_{\max}$ the longest horizon. The shared quantile state is θ_t ; source discrepancies/corrections are δ_t^j ; and (ζ_t, ψ_t) is the exogenous-adjustment block driven by covariates $\mathbf{x}_t$, such as precipitation, soil moisture, and the GDPC climate-index summary. Component discounting controls the evolution variances. The DQLM comparison in the Results panel is the conjugate AL-likelihood baseline; the selected exDQLM uses exAL and updates the non-conjugate σ, γ terms with VB-LD.

RESULTS

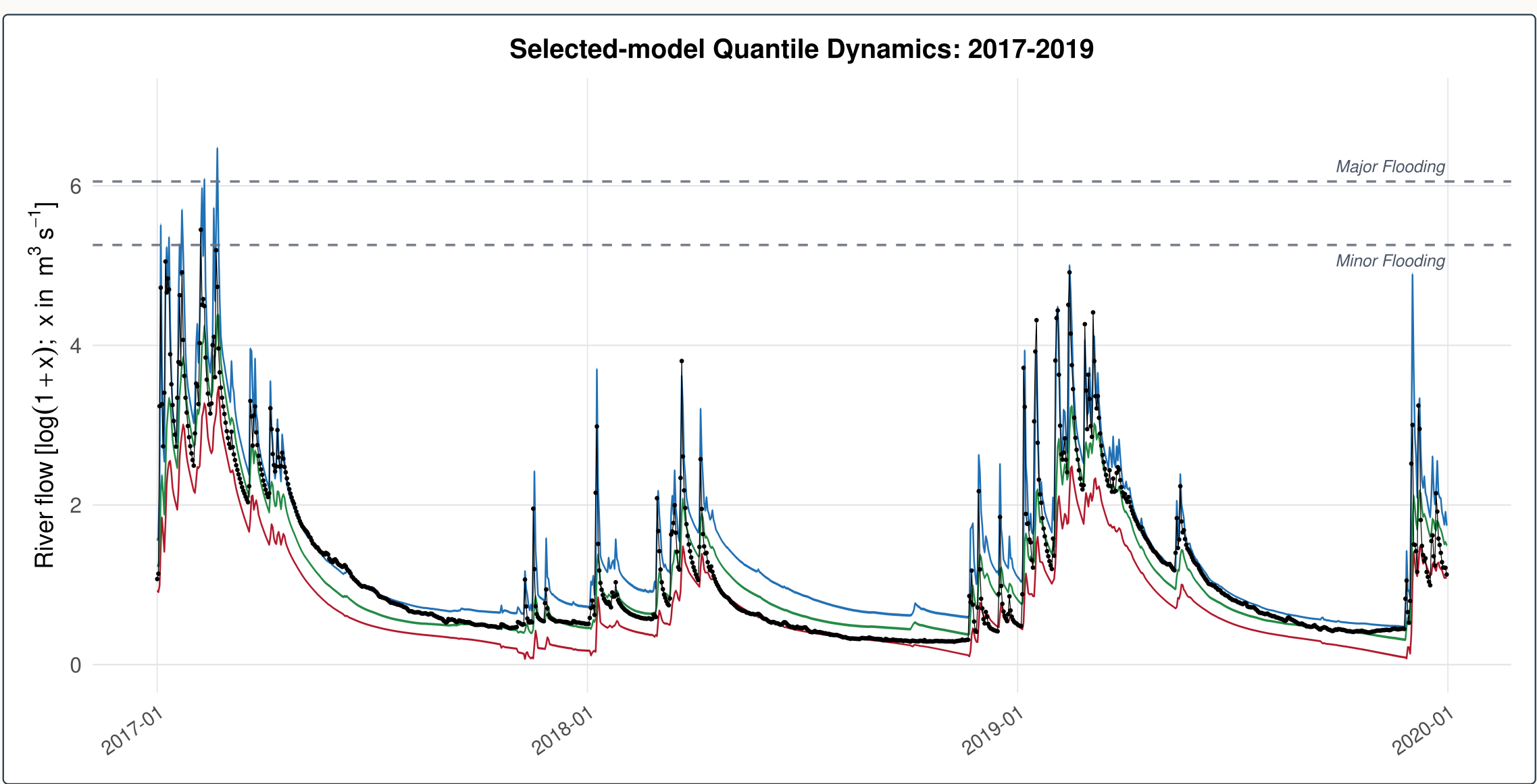
1–28-step-ahead forecast CRPS comparison



Values left of the raw GloFAS line indicate lower mean CRPS than raw GloFAS at that origin. DQLM denotes the conjugate AL-likelihood baseline; selected exDQLM denotes the exAL extension. Raw NWS is evaluated over common leads 1–8 in the horizon-matched tables, not mixed into this 1–28 lead GloFAS-supported benchmark.

FIT

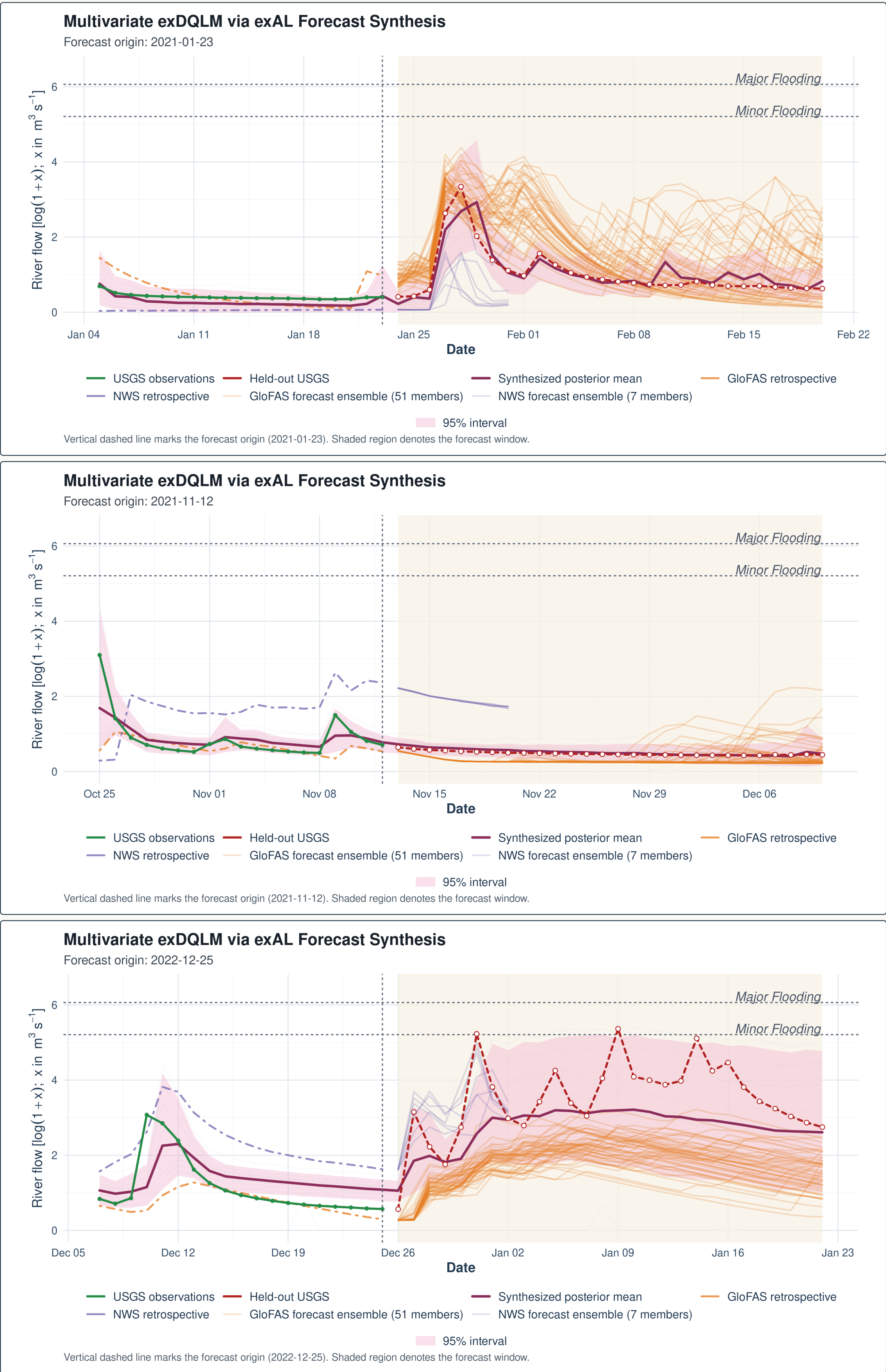
Wet-period quantile fit



Historical fit diagnostic. Selected exDQLM fitted quantile locations during the 2017–2019 wet period. The 5th, 50th, and 95th quantile fits are shown with observed USGS flow; wider upper-tail bands make winter high-flow behavior visible.

EXAMPLE

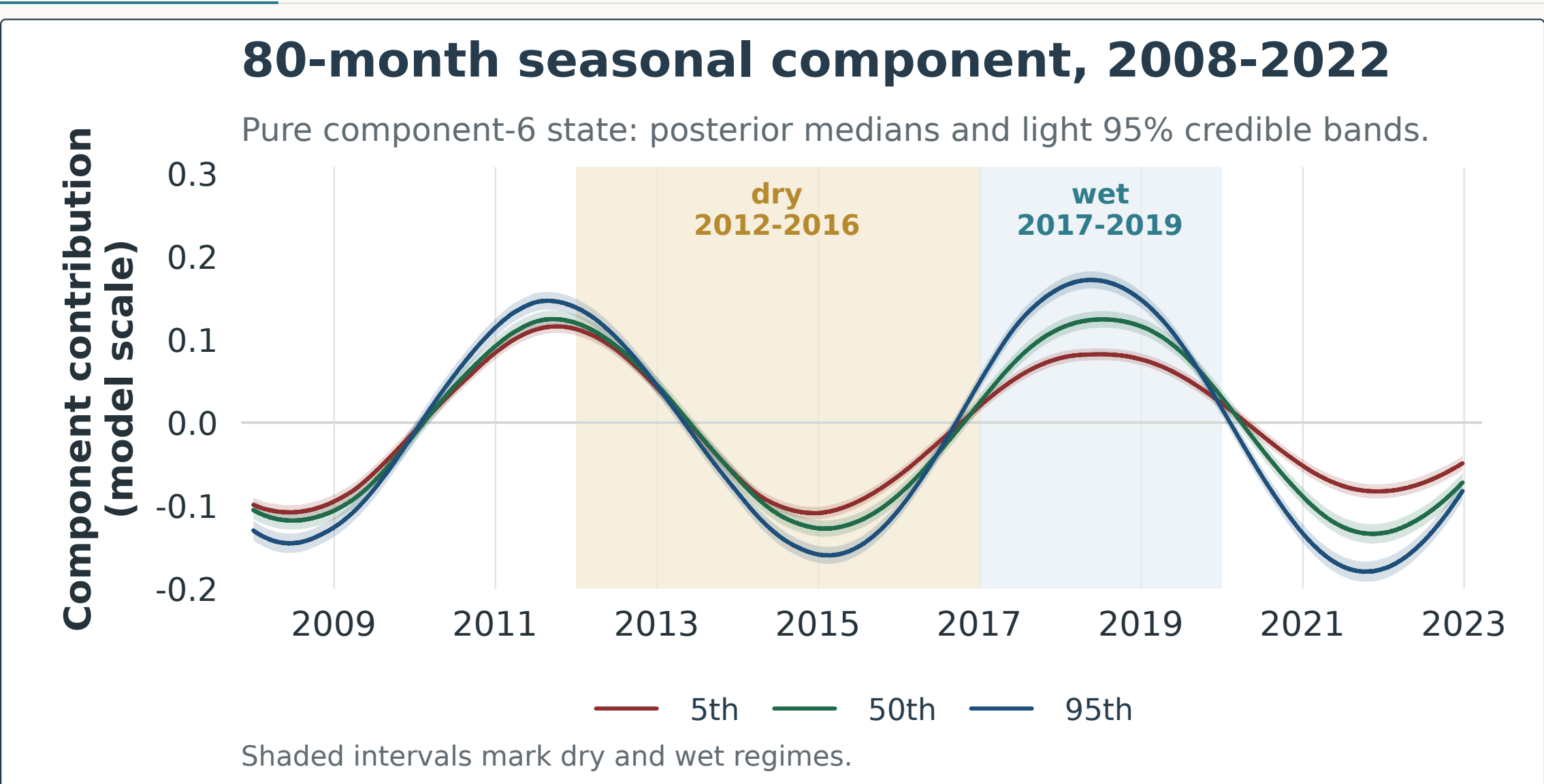
Three forecast-window examples



Origin-level illustrations. Three selected exDQLM forecast windows show how corrected quantile forecasts combine into a posterior predictive envelope after the cutoff. The panels are illustrative checks of behavior at individual origins; the aggregate comparison remains the five-origin CRPS benchmark.

DYNAMICS

80-month seasonal component



Retrospective diagnostic. From the selected exDQLM at the 2022-12-25 origin. Lines show the isolated 80-month state component, without the trend component added, for the 5th, 50th, and 95th target-quantile lanes over 2008–2022; light ribbons show 95% credible bands and shaded intervals mark dry and wet regimes.

TAKEAWAYS

Main takeaways

- Historical archives learn source-specific corrections before new forecast products are blended.
- The selected exDQLM extends the conjugate DQLM with exAL scale and skewness flexibility.
- Seven corrected quantile lanes synthesize one posterior predictive distribution.
- Selected exDQLM has the lowest 1–28-step-ahead CRPS at four of five origins; DQLM leads at the final origin.

